Working Title: Swim Team Scheduling and Performance Platform for University of Evansville Swimming & Diving

Client/Stakeholder: The primary client would be the University of Evansville Swimming & Diving coaching staff, particularly the head coach. Swimmers would also be major users and stakeholders of the system. I already have direct access to the coaching staff and swimmers, which will make communication, requirements gathering, feedback, and testing much easier. I have discussed the project and draft proposal with my coach, and he responded positively to the direction and current feature set.

Problem Statement: The University of Evansville Swimming & Diving coaching staff bear the cost of managing class schedules, team schedules, meet results, and team information through at least four different systems with no single place to view every component. This fragmentation is noticeable when scheduling practice times at the University of Evansville as the university does not have an allotted period of time for athletes to practice, which causes conflicts that the coaching staff must manually solve by comparing individual athletes' schedules. Can a unified swim-team scheduling and performance platform bring the core information from these separate sources into one system, identify 100% of conflicts in correctly entered schedules, and rank the lowest-conflict practice windows for the team in 10 seconds or less?

Background/Literature Review:

- Swimming in Data (2024): This research shows that competitive swimming is becoming more data-driven and that performance data can be extremely useful for swimmers/coaches.
- NCAA Division I Time-Demands Research (2024): Shows that the balancing of academics, athletics, and competition is a real issue for Division I athletes and coaches, supporting the need for the project.
- Streamlining Performance Prediction (2024): This source showcases that swimming performance can be analyzed through KPIs, supporting the analytical performance side of the project.
- Predicting Future Stars (2024): Demonstrates how athlete data can be used to visualize athlete development and progression over time.
- Integrated Training Load and Injury/Illness System (2023): Showcases that systems can be used in competitive swimming while also highlighting how important usability, privacy, and stakeholder buy-in truly are.
- Training Load Monitoring in Competitive Swimming (2022): This source shows that many swimming programs use some sort of training monitoring and find it to be useful, but avoid its use if data entry is too burdensome.
- Swimming Coaches' Professional Development and Training Practices (2023): Displays that performance analysis technology should be designed around coaching needs rather than just displaying data.

Existing Systems/Prior Work:

- SwimCloud: This system has swimmer profiles, meet results, and performance progression. It is a good example of the performance side of the project, but it does not have anything to do with academic schedules and the conflict-management portion of my project.
- Teamworks: This system has academic scheduling, calendars, classes, practices, and communication. It is a great example for the scheduling side, but it does not include anything about performance.
- ARMS: Overlaps with many of Teamworks' features but is more specifically focused on collegiate athletics administration and NCAA compliance. Good example of an app that is focused on what I want.
- Commit Swimming: Offers many features tailored to swimming such as training history, results, analytics, and scheduling. This is one of the best examples of what I would like to build, but it is completely tailored to club swimming not college swimming, which is very different.

- SwimTopia: Combines team management, calendars, communication, and results, but does not focus on academic-schedule conflicts and practice optimization for college swimmers.

Gap/Need Being Addressed: Existing products solve pieces of the problem, but I have not found a solution that includes all the pieces and is specifically tailored for a collegiate swimming team. There is no system I could find that combines academic and team scheduling, conflict detection, swim-specific performance progression, goals, and optional training information. My project will be specifically tailored and cover the needs of the University of Evansville Swimming and Diving team rather than trying to replace all of the larger systems that are pieces of the puzzle.

Proposed Solution: I would develop a web-based platform for collegiate swimmers and coaches that combines team and academic scheduling, conflict detection, performance tracking, goals, and other useful team information into one system. The initial system would be designed around the needs of UE Swimming & Diving, with the exact requirements refined through further discussion with the coaching staff and swimmers.

Core Features:

1. Unified Scheduling and Calendar - Display class schedules, practices, meets, travel, and other team activities in one place.

2. Conflict Detection and Coach Availability Dashboard - Automatically identify conflicts and allow coaches to see team availability and determine practice times that create the fewest conflicts.

3. Performance Tracking and Analytics - Store race results and personal-best times and display progress by career, season, semester, meet type, and event.

4. Athlete and Team Goals - Allow swimmers to create individual/team goals, share selected goals, and support teammates.

5. Optional Practice Tracking - Allow swimmers to record practice times, sets, notes, and other useful training information.

Preliminary Technical Approach: I will most likely develop the project as a full-stack web application using a frontend framework such as React, Python for backend functionality, and a relational database for schedules, swimmer information, results, goals, and other data. Algorithms would be used for schedule conflict detection and team availability, while data visualization tools could display swimmer performance and progress. The exact technologies may change as the project requirements become more defined. The system should not depend on direct integration with external platforms to function; manual entry or file-based imports could be used if APIs or automated integrations are unavailable.

Feasibility and Risks: I believe the project is feasible because I already have direct access to the client and users, understand the problem firsthand, and can develop the system in separate modules using technologies that are realistic for me to learn and use. The two largest risks are scope and data access/integration. There are many possible features, so the five identified core features will define the main scope of the project. Performance or scheduling data may also be difficult to import automatically from existing systems, so the platform should support manual or file-based data entry if necessary. No specialized hardware or significant project funding should be required.

Responsible Computing Concerns: The main concerns are privacy and security for this project. The system could potentially contain class schedules, performance data, goals, and other information about student-athletes. I would have to ensure that a coach versus an athlete have different access to certain features, and I would also have to avoid using performance statistics in ways that create unfair comparisons between athletes.

Success Criteria: The project would be successful if the finished prototype allows coaches and swimmers to manage schedules and performance information in one location, identifies 100% of conflicts in correctly entered schedules, and ranks low-conflict practice windows within 10 seconds. The system should also correctly store and display swimming performance information. Usability testing and feedback from the coaching staff and swimmers would be used to determine whether the platform is organized, useful, and something they would want to continue using.

Revision Summary: This revised proposal is now narrower and better clarifies the project scope based on feedback as well as research. I updated the problem statement as that was something we worked on in class and highly improved, making it more measurable. I heavily expanded on the background and existing systems in this proposal giving more evidence as to why this project is important. I clearly identified the gap in existing solutions that my project addresses and also finalized five core features after discussing the project with my client.

References:

Scholarly / Technical Sources

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Santosa, J. C., & Sagas, M. (2024). *Understanding the differences in U.S. NCAA Division I coaches and athletes' preferences on time demands measures.* Cogent Social Sciences. https://www.tandfonline.com/doi/full/10.1080/23311886.2024.2302218

Staunton, C. A., et al. (2024). *Streamlining performance prediction: Data-driven KPIs in all swimming strokes.* BMC Research Notes. https://link.springer.com/article/10.1186/s13104-024-06714-x

Existing Systems / Products

ARMS. *ARMS / Teamworks Compliance.* https://my.armssoftware.com/arms/

Commit Swimming. *Commit Swimming.* https://www.commitswimming.com/

SwimCloud. *SwimCloud.* https://www.swimcloud.com/

SwimTopia. *SwimTopia*.
https://www.swimtopia.com/

Teamworks. *Teamworks Hub*.
https://teamworks.com/hub